ID	Title	Epic	User Story	Priority (MoSCoV	Status	Acceptance Criteria	Functional Requirements	Non-Functional Requirements	Original Estimate	Actual Effort (In days)
1	Hyperspectral Data Extraction	Phase 1: Data Engineering	As a researcher, I need to extract raw spectra and labels from .rda files into a CSV so the neural network can ingest them.	Must Have	Done	.rda files parsed into a 256-feature Pandas DataFrame with intact string labels.	Script must parse R data objects and dynamically map polymer text labels to continuous feature columns.	Parsing script must be optimized (e.g., using compiled C-backend via pyreadr) to handle large datasets efficiently.	2	3
2	Reflectance Calibration	Phase 1: Data Engineering	As a data engineer, I need to calibrate raw intensity against the white standard to calculate true reflectance.	Must Have	Done	Output feature matrix represents true reflectance without dimension mismatch errors.	System must apply matrix division (Raw / White Standard) and plot a calibrated 1D sample curve.	High computational accuracy for floating-point arithmetic across 256 wavelengths.	1	1
3	1D-CNN Development	Phase 2: Spectral Classification	As an ML engineer, I need to build and train a 1D-CNN to classify chemical polymer types based on their spectral absorption curves.	Must Have	Done	PyTorch model compiles, executes, and successfully trains over 25 epochs.	Model must accept input shape of (Batch, 1, 256) and output probability logits for PE, PP, PET, and PS.	Inference execution time must be under 10ms per sample for real-time application.	4	4
4	Overfitting & Robustness Audit	Phase 2: Spectral Classification	As a researcher, I need to validate the model's accuracy so I can prove it generalizes well to unseen samples.	Should Have	Done	Model achieves >95% validation accuracy using Stratified 5-Fold Cross-Validation.	System must evaluate test splits and output a Classification Report and Confusion Matrix heatmap.	Model training and evaluation must be fully reproducible (utilizing fixed random states).	2	2
5	Morphological Preprocessing	Phase 3: Image Processing	As a computer vision engineer, I need to batch-resize and normalize microscopic images to 224x224 so they are ready for the 2D-CNN.	Must Have	To Do	All images in train and valid folders are normalized and loaded into PyTorch DataLoaders.	Script must apply RGB channel normalization and standard center-cropping/resizing transformations.	High throughput scalability to process batches of 32 high-resolution images simultaneously without memory leaks.	3	0
6	2D-CNN Transfer Learning	Phase 3: Image Processing	As an ML engineer, I need to fine-tune a pre-trained ResNet18 model to classify the physical shapes of microplastics (fibers, fragments, films).	Must Have	To Do	PyTorch ResNet18 model successfully modifies the final fully connected layer and trains for 15 epochs.	System must apply transfer learning techniques (freezing/unfreezing layers) and optimize using CrossEntropyLoss.	GPU acceleration (CUDA) must be utilized to maintain reasonable training times for image data batches.	4	0
7	Morphological Model Evaluation	Phase 3: Image Processing	As a researcher, I need to evaluate the 2D-CNN's performance using a confusion matrix to ensure it accurately distinguishes between physical shapes.	Should Have	To Do	A visual confusion matrix and classification report are successfully generated for the validation image set.	Script must perform inference on the validation DataLoader and calculate precision, recall, and F1-scores per class.	Evaluation metrics must be statistically rigorous and generated without data leakage from the training set.	2	0